

Conceptual Design for a BSL-3 Suitable Stage-Top Incubator



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Chapter 1

Introduction

1.1 Background and Summary

Long term Live-Cell imaging is a very useful research tool when determining long term behavior of cell cultures in various situations. Commercial solutions are unable to provide adequate containment for the use within a Bio-Safety Level 3 (BSL-3) and often require the use of a separate containment enclosure, which introduces additional costs and complexity.

This project focuses on the conceptual design of a low-cost Stage-Top Incubator that is capable of being utilized within a BSL-3 laboratory, without the requirement for additional containment.

1.2 Scope and Limitations

This project is limited to the following:

- Preliminary Research
- Temperature Control
- Compatibility with Multi-Well microtiter plates

This project is limited to compatibility with a BSL-3 Lab and does not consider higher containment requirements. The environmental control aspects of the incubator are limited to just temperature control for the preliminary conceptual design. The initial design is limited to compatibility with a multi-well microtiter plate.

Chapter 2

Research

The conceptual design began with assessing existing commercial products to identify the key environmental control specifications required for the incubator to function correctly. Additionally, this was useful in identifying why the commercial products are incompatible with a BSL-3 lab when used without a separate external biological containment mechanism.

2.1 Zeis/Pecon

The Zeis/Pecon stage top incubator utilities four modules to control the environmental conditions within the incubator stage.

The Pecon Temperature Module S[1] regulates the temperature of the incubator. It has four 8-pin connectors that are used for temperature control of heated components such as the Incubator lid, base, microscope objective and humidifier device. The temperature module is the base module for the entire incubation system, providing electrical power and communications to all connected devices. It directly connects to the CO₂ module to provide power, communication and makes a connection to the gas circulation system to heat the gas flow provided to the incubation chamber. The temperature module requires an external device such as a PC or compatible microscope to operate as it does not have any control panel. The module maintains culture temperature by monitoring and controlling the temperature of the incubator base, lid and optionally the microscope objective. The module infers the culture temperature using predefined temperature relations, or by performing a pre-experiment calibration whereby an identical set-up is placed in the chamber with the included culture temperature sensor to tune the required chamber temperatures.

The Pecon CO₂ Module S[2] attaches to the temperature module for electrical power and communication, and provides CO₂ concentration control to the airflow passed into the incubation chamber. The CO₂ module circulates gas throughout all the connected units and mixes the gas before pumping the resultant mixture to the incubation chamber. The module contains a CO₂ sensor which monitors the concentration of CO₂ in the gas mixture, and controls the release of pure CO₂ into the mixing chamber in order to maintain the desired CO₂ concentration. Due to the method of operation, it requires a >95% purity CO₂ supply.

The Pecon O₂ Module S[3] attaches to the CO₂ module for electrical power and communication. This module functions similarly to the CO₂ module, however it regulates the concentration of O₂ present in the gas mixture by controlling the release of a pure N₂ supply into the air mixture to displace the O₂ present. This method of O₂ displacements means that the O₂ module is only capable of providing hypoxia conditions by lowering O₂ concentration. Due to the method of operation, it requires a >99% purity N₂ supply.

Both the CO₂ module and O₂ module are supplied with a humidifying bottle which functions by bubbling the the gas airflow thorough distilled water to raise the relative humidity (RH). The Pecon Heating Device Humidity S1[4] is an optional module which connects to the Temperature module to heat the humidifying bottle to increase the RH and maintain adequate gas temperature provided to the incubation chamber

The incubator stage includes a heated glass topped lid[5] and a heated insert which accepts mounts for the different dishes used in microscopy. The heated insert directly heats the culture medium through physical contact, and the heated lid provides radiative heating and condensation prevention by heating the glass panel directly. The insert supports perfusion experiments utilising two ports which allow tubes to be run to the culture

2.2 Ibidi

The Ibidi stage top incubator functions similarly to the Zeis/Pecon module, however it makes use of only 3 modules to maintain environmental conditions.

The Heating System Silver Line module regulates the temperature of the incubation chamber stage and the microscope. The heating of the gas mixture is subjected to the gas regulator module.

The Gas Incubation System comes in two different models, the CO₂ - Silver Line and the CO₂/O₂ - Silver Line. The difference between these two systems is the former is only capable of regulating CO₂ concentration, where as the later has the ability to regulate O₂ concentration as well as CO₂ concentration. The gas incubation system is responsible for heating the gas mixture and the humidifying column to the appropriate temperature.

The Humidifying system works by mixing airflow directly from the Mixing module and from the humidifying column, which bubbles a separate airflow through heated water to obtain approximately 100% RH air. The gas mixer module then controls the mix ratio of dry gas to 100% RH gas to obtain a controlled humidity flowing into the incubation chamber.

The Ibidi incubation chamber comes in two versions, one version supports culture mediums with Petri dishes and slides, the second version supports multi-well plates. The Petri dish/Slide version of the chamber supports perfusion experiments using two ports which allow tubes to be run to the culture. Both versions support direct medium temperature monitoring which allow for more accurate temperature control by directly measuring the temperature of the culture medium.

The Ibidi system functions in a very similar fashion to the Pecon system, with the addition of being able to control the relative humidity in the chamber, rather than a forced approximate 100% relative humidity. This system still poses the same problem wherein the airflow is designed to passively exhaust from the incubation chamber, as shown in [Figure 2.1](#), and therefore does not have a seal that meets BSL-3 containment standards.

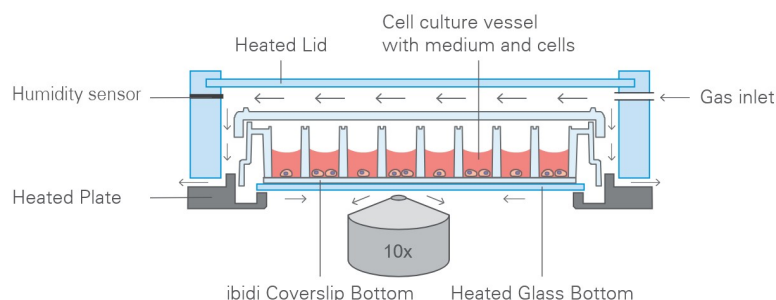


Figure 2.1: Diagram Showing the Airflow Path Within an Ibidi Multi-Well Incubator Chamber [6]

2.3 Tokai-Hit

The Tokai-Hit system functions similarly to the Pecon and Ibidi systems however it utilises a single module to regulate the environmental conditions for temperature and CO₂. Humidity is controlled by a heated water bath inside the chamber, and O₂ control is achieved by utilising an optional module which then overrides all gas control.

The Tokai-Hit STXG/STXF temperature and CO₂ controller regulates both temperature and CO₂ concentration in the incubation chamber. The controller is capable of regulating the temperature of the top-lid heater, water bath heater, stage heater and the microscope objective heater. Additionally it is capable of regulating the airflow into the chamber depending on the model and air supply.

The incubator chamber uses a single design that is compatible with all major culture dishes including Petri dishes, slides and multi-well plates. This chamber supports perfusion experiments, drug administration and direct culture temperature measurement via access ports designed into the lid.

This system still poses the same issue as the aforementioned chambers, which prevents this chamber from forming a completely sealed environment suitable for a BSL-3 lab.

2.4 Oko-Lab

The Oko-Lab utilises three modules to regulate environmental conditions within the incubation chamber

The Oko-Lab H301-T-UNIT-BL-PLUS [7] unit is responsible for maintaining the thermal conditions within the chamber enclosure. It operates by monitoring and controlling the temperature of the chamber base, lid and optionally the microscope objective. This in turn heats the sample to the desired temperature. This heating unit is capable of sample temperature feedback utilising a thin thermocouple that can be placed in distilled water near the sample to obtain more accurate sample temperature control. Otherwise the unit is capable of using pre-sets/calibrations to infer sample temperature based on enclosure temperatures. The heating unit can be operated via a computer using available software or using optional touch screen accessory. The heating unit is capable of interfacing with the Oko-Lab Bold Line Gas Controllers.

Chapter 3

Feasibility Analysis

3.1 Traceability Analysis

Show how the requirements, specifications and testing procedures all link, [Table 3.1](#) is provided.

Table 3.1: Requirements Traceability Matrix

#	Requirements	Specifications	Acceptance Test
1	UR01	SP01	AT01
2	FR01	SP02,SP05	AT03

3.1.1 Traceability Analysis 1

3.1.2 Traceability Analysis 2

3.2 Requirements

The requirements for a micromouse power module are described in [Table 3.2](#).

Table 3.2: User and functional requirements of the power subsystem.

Requirement ID	Description
UR01	
FR01	

3.3 Specifications

The specifications, refined from the requirements in [Table 3.2](#), for the micromouse power module are described in [Table 3.3](#).

Table 3.3: Specifications of the sensing subsystem derived from the requirements in [Table 3.2](#).

Specification ID	Description
SP01	User

3.4 Acceptance Testing Procedures

Table 3.4: Subsystem acceptance tests

ID	Description	Testing Procedure	Pass/Fail Criteria
AT01	Powers on	• •	
		• •	

Chapter 4

Subsystem Design

4.1 Design Decisions

4.2 Final Design

The following design...



Figure 4.1: Schematic

4.3 Failure Management

Table 4.1: CAPTION

Name	Description

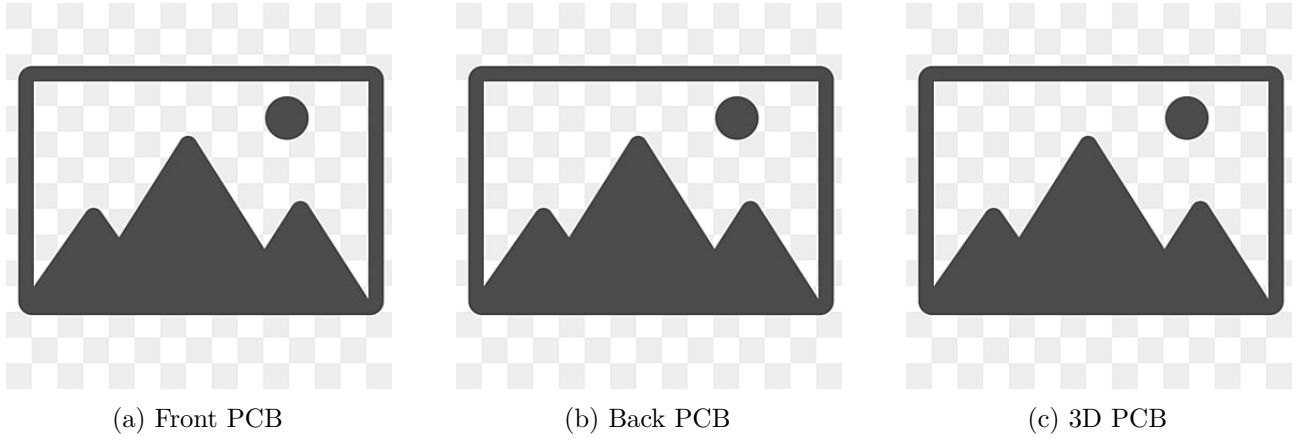


Figure 4.2: PCB

4.4 System Integration and Interfacing

To integrate the subsystem with the rest of the system

Table 4.2: Interfacing specifications

Interface	Description	Pins/Output
I001	SparkFun 9DoF IMU Breakout board to STM for data transfer (SPI)	• MISO: Breakout MISO* to STM PB14 • MOSI: Breakout MOSI* to STM PB15 • SCLK: Breakout SCLK* to STM PB13 • CS: Breakout CS* to STM PB12 (GPIO) • Power: Breakout 1V8-5V5* to STM 3V • GND: Breakout GND* to STM GND
I002	STM to FTDI for data transfer to computer (UART)	• STM PA2(RX) to FTDI RXD • STM PA3(TX) to FTDI TXD
I003	FTDI to computer for data transfer from STM	• Assuming the FTDI is connected to a Micro B USB port (which can be used to connect to a USB cable to connect to a computer), output via cable

Chapter 5

Discussion and Results

Chapter 6

Conclusion

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6.1 Recommendations

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